Edge AI-Based Defect Detection in White Pepper (Piper nigrum L.) Using CLAHE-Based Pre-processing and YOLO

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Abstract—The quality sorting of white pepper is essential in agriculture, especially to support Indonesia's position as a major pepper exporter. Existing defect detection models face limitations under poor lighting conditions, affecting the accuracy of automated sorting. This study proposes a YOLOv8-based defect detection model integrated with the Edge AI device NVIDIA Jetson Orin Nano. To enhance image contrast and improve detection under varying lighting, CLAHE pre-processing is applied. The dataset includes images of white pepper classified into two categories: normal and defective. The proposed system performs real-time detection with low latency, leveraging the Jetson Orin Nano's on-device processing capabilities for efficiency and mobility. A Streamlit-based interface is also developed for real-time visualization. Evaluation results demonstrate improvements across all metrics when using CLAHE, achieving accuracy, precision, recall, specificity, and F1-score of up to 99%, and an mAP50-95 of 82%. Additionally, the IoU scores exceeding 90% indicate a high level of detection accuracy. The results aim to support AI-driven solutions in agricultural quality control.

Keywords— White Pepper, YOLOv8, NVIDIA Jetson Orin Nano, CLAHE, Defect Detection.

I. INTRODUCTION

Indonesia is an agrarian country with significant potential to expand its agricultural exports, particularly in the plantation subsector. One of the key export commodities in this subsector is pepper, which plays an important role in the national economy as both a source of foreign exchange and a livelihood for farmers. Pepper is one of Indonesia's leading spice commodities within the plantation subsector and has contributed significantly as the largest export commodity in the spice category. According to data from the Directorate General of Plantations (Ditjenbun, 2023), Indonesia's pepper exports in 2020 reached 58,378 tons. As one of the world's major producers and exporters of pepper, Indonesia competes with other leading pepper-producing countries such as Vietnam and Brazil [1].

In addition, the global price of pepper has increased in line with population growth, which has driven higher demand. Indonesia's pepper exports showed a declining trend during the period of 2015–2018, with an average decrease of 34.02%. However, in 2015, there was a significant surge in export value, increasing by 69.30%. In 2018, the export value of pepper was recorded at USD 152.47 million, with a trade surplus of USD 147.9 million, representing a 36% decrease compared to the previous year. The highest surplus was recorded in 2015, with a trade balance reaching USD 535 million [2].

In 2019, Indonesia exported approximately 21,452 tons of pepper, accounting for 41.44% of the total global export volume. China was the main export destination, receiving nearly 7,000 tons, or about 12.92% of Indonesia's total pepper exports [3].

To meet this growing global demand, ensuring consistent post-harvest quality is essential. To ensure high quality, pepper seeds undergo soaking in clean water for 7–10 days to soften the outer layer, followed by sun-drying for 3–7 days. Once dried, seeds are manually sorted to separate clean white seeds from defective ones [4]. Technology, particularly camera-based systems, is recommended to automate sorting based on shape and size, improving efficiency and ensuring uniform, high-quality output.

To enhance the reliability and efficiency of the sorting process, a deep learning approach using YOLOv8 (You Only Look Once) with a Convolutional Neural Network (CNN) architecture will be employed to detect defects in white pepper. YOLOv8, based on CNN, enables real-time object detection by dividing images into grids and assessing the likelihood of defects in each grid. YOLOv8 is selected due to its superior compatibility and performance when deployed on Edge AI devices, such as the Nvidia Jetson Orin Nano, which is utilized in this research. Compared to other object detection models like Faster R-CNN or SSD, YOLOv8 offers significantly lower latency and faster inference while maintaining high accuracy. Additionally, Contrast Limited Adaptive Histogram Equalization (CLAHE) pre-processing will be applied to improve image contrast, particularly in low-contrast conditions.

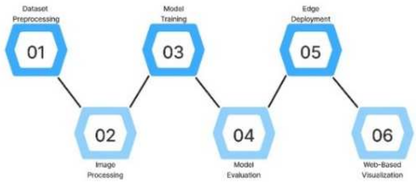


Fig. 1. Pipeline of the Proposed System

The Contrast Limited Adaptive Histogram Equalization (CLAHE) pre-processing method is a technique used to enhance image contrast, particularly in areas with significant differences in brightness or grayscale values [5]. While several contrast enhancement techniques exist, such as global histogram equalization and Retinex-based methods, CLAHE was selected for its ability to improve local contrast without significantly amplifying noise. Compared to traditional methods, CLAHE offers adaptive adjustment across localized regions, making it particularly effective for detecting subtle defects in pepper seeds under non-uniform lighting conditions.

The trained YOLOv8 model will be implemented on the NVIDIA Jetson Orin Nano Edge AI device. This implementation allows for real-time detection of white pepper with low latency, supporting the efficient sorting of white pepper in the field. Edge AI is preferred over traditional cloud-based systems, as the latter may struggle to provide real-time responses across all connected devices without sacrificing user experience due to latency or inaccuracies. Edge AI, offering low latency, high mobility, and the ability to handle multiple nodes across distant locations, is the ideal solution for this research [6]. Additionally, the Streamlit framework will be used to visualize the real-time defect detection process, enabling users to directly detect defects in white pepper.

II. Method

The proposed system is designed to operate through a series of systematic stages, as illustrated in Figure 1. The initial stage involves processing a dataset obtained from previous research, which contains images of white pepper categorized into two classes: normal and defective. Following this, image enhancement is performed using the Contrast Limited Adaptive Histogram Equalization (CLAHE) technique to improve visual details and emphasize significant features relevant to the classification process. Both the original and CLAHE-enhanced images are used as training data for deep learning models assigned to perform object detection. These models are trained and subsequently evaluated using various performance metrics, including precision, recall, specificity, F-score, mean Average Precision (mAP), and Intersection over Union (IoU), in order to identify the model with the highest accuracy and efficiency in distinguishing between normal and defective pepper.

Upon completion of the evaluation process, the best-performing model is deployed on the NVIDIA Jetson Nano Edge AI device to support real-time inference with optimal computational efficiency. This implementation aims to demonstrate that the system can run directly on low-power devices without compromising detection speed or accuracy. In the final stage, the system is integrated into an interactive web-based interface developed using Streamlit. Through this interface, users can upload white pepper images directly and

Fig. 2. CLAHE Result

instantly receive classification results, along with the corresponding inference time. Thus, the entire process from image input, visual enhancement, and object detection to the presentation of classification results can be carried out in a fully integrated and user-accessible manner.

A. Dataset Pre-processing

This research employs a white pepper image dataset sourced from prior studies, accessible via Roboflow [7]. The dataset comprises a total of 1,269 images, all of which have been annotated, yielding 2,388 instances of the normal class and 1,412 instances of the defective class. It is organized into three subsets: training, validation, and testing, each ensuring a balanced distribution of classes. All images are standardized to a resolution of 224 × 224 pixels, aligning with the input specifications of the deep learning model utilized in this study, specifically YOLOv8m. Specifically, the training set consists of 901 images, containing 1,674 instances of the normal class and 1,050 instances of the defective class. The validation set comprises 227 images with 441 instances of the normal class and 281 instances of the defective class, while the test set includes 141 images, containing 273 normal and 81 defective instances. This well-structured, balanced, and high-quality dataset serves as a critical foundation for both the training and performance evaluation of the proposed deep learning model.

B. Image Processing

To enhance the visual quality of the input images, a pre-processing pipeline was developed with CLAHE (Contrast Limited Adaptive Histogram Equalization) as its core component. Although CLAHE effectively improves local contrast, it may introduce undesirable artifacts and amplify noise in certain regions. To address these limitations, several complementary pre-processing techniques were applied sequentially.

First, gamma correction was performed to brighten the image by adjusting pixel intensity using a gamma transformation [8]. CLAHE was applied to localized regions to enhance contrast. Subsequently, bilinear interpolation was used to blend the enhanced patches and minimize visible edge artifacts [9]. Next, a denoising step was carried out to suppress image noise while preserving important structural details. Non-local means (NLM) is one of the most representative patch-based denoising algorithms, which utilizes the redundancy and similarity among various parts of the image to effectively reduce noise without compromising image structure [10]. Finally, unsharp masking was employed to sharpen image edges using Gaussian blurring and contrast enhancement, Gaussian blur was applied to generate a low-pass filtered image, which served as the basis for high-pass filtering aimed at enhancing fine details [11]. This pre-processing combination aims to balance contrast enhancement, noise suppression, and edge preservation for optimal image quality.

C. YOLO (You Only Look Once)

The YOLO (You Only Look Once) algorithm is a deep learning-based object detection method that applies a single neural network to the entire image [12]. To enable real-time object detection, YOLO utilizes a Convolutional Neural Network (CNN) to process visual data. The detection process begins by taking an input image, which is then divided into an S × S grid to generate bounding boxes. The system predicts class probabilities based on the confidence level in identifying an object [13]. Each bounding box includes five prediction parameters: the x and y coordinates of the center, the width (w), the height (h), and the confidence score. The confidence score reflects the system's certainty regarding the presence of an object within the bounding box, as well as the accuracy of the grid cell in localizing the object [14].

The main advantages of the YOLO algorithm include high speed, good efficiency, the ability to detect multiple classes simultaneously, and a flexible architecture. This algorithm can predict the presence of objects in a short amount of time, process images efficiently in parallel, detect various object classes concurrently, and be adapted to different application needs [15].

YOLOv8, introduced in 2020, improves upon previous versions with a more efficient backbone, multi-scale predictions, and a new anchor system. Its architecture includes a backbone using Feature Pyramid Network (FPN) for feature extraction, a neck with Cross Layer Connection (CLC) for refinement, and a head that generates bounding boxes, class scores, and accuracy for object detection [16].

D. Model Training

In this study, the YOLOv8m architecture was employed to develop the object detection model. This architecture was selected due to its high capability as CNN model in detecting multiple objects in real-time with both efficiency and accuracy. The training process was conducted using the PyTorch framework by utilizing the YOLOv8 model developed by Ultralytics, and the entire training was carried out on the Google Colaboratory cloud platform, which offers flexible and efficient GPU resources.

The dataset used for training consists of two classes of images, namely white pepper in normal condition and white pepper with defects. These images were divided into two separate datasets one without any pre-processing and the other enhanced using the CLAHE technique. Both sets of images were resized to 224×224 pixels. To increase the diversity and volume of the training data, several data augmentation techniques were applied, including horizontal and vertical flipping as well as image rotation, with the goal of enhancing the model's generalization ability to real-world image variations.

E. Evaluation Model

A fundamental instrument for assessing classification models is the confusion matrix, which illustrates the interplay between accurate and inaccurate predictions. This matrix underpins the computation of essential performance metrics, including accuracy, precision, recall, and specificity, thereby offering a comprehensive insight into the model's classification efficacy. The confusion matrix consists of four main components: True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN), each representing a specific outcome of the model's prediction compared to the actual class.

Additionally, the F1-score, representing the harmonic mean of precision and recall, is frequently utilized to gauge the balance between these two metrics. The calculation of the F1-score is calculated using the following equation [17]:

$$F_1 = \left(\frac{recall^{-1} + precision^{-1}}{2}\right) = 2 \cdot \frac{precision \cdot recall}{precision + recall} \quad (1)$$

Moreover, this study also considers mean Average Precision (mAP) and Intersection over Union (IoU) as evaluation metrics. The mAP values are directly obtained from the YOLOv8 model evaluation, while the IoU scores are manually calculated using the standard formula to further assess the model's performance in object detection and classification tasks. The IoU is computed as follows:

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} \quad (2)$$

F. EDGE AI Configuratioon

Edge AI enables artificial intelligence algorithms to be executed directly on local hardware devices. This technology allows devices to independently process generated data and make decisions without the need for a continuous network connection [6]. By decentralizing computation from the cloud to the device level, Edge AI significantly improves responsiveness and reliability in time-sensitive applications. As a result, Edge AI offers several advantages, including reduced latency, enhanced data privacy, bandwidth efficiency, and the ability to operate in real-time under limited connectivity conditions. One of the notable Edge AI devices is the NVIDIA Jetson Orin Nano, an edge computing platform provided by NVIDIA, which is equipped with a high-performance Graphics Processing Unit (GPU) to accelerate AI processing at the edge [18]. The complete specifications of this variant are presented in Table II.

G. Web-Based Visualization

The Streamlit platform offers an interactive interface that allows users to upload data, load AI models, adjust hyperparameters, test adversarial attack robustness, train and evaluate models, and apply mitigation techniques such as adversarial training and defensive distillation [20]. In this study, Streamlit is utilized as a user interface for detecting defects in white pepper seeds using a pre-trained YOLOv8m model.

III. Results and Discussions

This research details the application of Edge AI utilizing Streamlit for the identification of defects in white pepper. The defect detection is executed through the YOLOv8m deep learning framework, chosen for its exceptional performance

TABLE I.

HYPERPARAMETER SETTINGS FOR MODEL TRAINING

Parameter	Without CLAHE	With CLAHE
Epoch	50	50
Batch Size	16	16
Learning Rate	0.0001	0.0001

TABLE II.

SUMMARY OF JETSON ORIN NANO DEVICE [19]

Specification	Description
Name	Jetson Orin Nano 8G
Released Time	January 2023
AI Performanca	40 TOPs
GPU Detaik	1024-core NVIDIA Ampere GPU with 32 Tensor Cores
CPU Detail	6-core Arm® Cortex®-A78AE v8.2 64-bit, 2.2 GHz
Power	7-15 W
Memory	8GB 128-bit LPDDR5, 2133 MHz frequency

as documented in earlier research. The model underwent training on a dataset consisting of two categories of images: intact white pepper and those exhibiting defects. These images were categorized into two distinct datasets—one that did not undergo CLAHE pre-processing and another that incorporated CLAHE-based pre-processing.

A. *Implementation of Pre-processing*

To enhance object detection performance, this study employs CLAHE (Contrast Limited Adaptive Histogram Equalization) as the primary pre-processing technique to improve the image quality of pepper seeds. In addition to CLAHE, gamma correction is applied to brighten the images, denoising is performed to suppress image noise while preserving details, and unsharp masking is used to sharpen image edges. Table III presents a visual comparison between the original and preprocessed images for both normal and defective categories.

In images of both normal and defective pepper seeds, the enhancement effect of CLAHE-based pre-processing is relatively subtle in the background. Before pre-processing, the background exhibits slight noise, which is completely removed after pre-processing. The most noticeable improvement lies in the object details, as CLAHE enhances the contrast within the object, resulting in clearer and more defined features. By combining several enhancement techniques chiefly CLAHE the resulting images exhibit clearer visual distinctions between preprocessed and raw seed areas. This improvement is vital for accurate object detection, as it enhances feature clarity and enables the deep learning model to better differentiate between normal and defective white pepper seeds.

B. *Evaluation of Model Performance*

The classification performance was evaluated on two trained models: one using the original dataset without CLAHE and the other using the CLAHE-enhanced dataset. The evaluation employed seven key metrics: Accuracy, Precision, Recall, Specificity, F1 Score, mAP Score, and IoU Score.

TABLE III.

CLAHE PRE-PROCESSING RESULTS ON PEPPER IMAGES

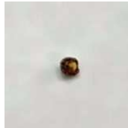
Normal Class		Defective Class	
Before	After	Before	After
			
			

TABLE IV.

EVALUATION METRIC RESULTS

Metric	Without CLAHE	With CLAHE
Accuracy	98%	99%
Precision	98%	99%
Recall	98%	99%
Specificity	97%	99%
F1-Score	98%	99%
mAP50-95	79%	82%

As shown in Table IV and Figure 3, the application of CLAHE improved the model's performance across all evaluation metrics. For the model trained on the dataset without CLAHE, the scores for accuracy, precision, recall, and F1-score were 98%, with a specificity of 97% and an mAP50-95 of 79%. In contrast, the model trained on the dataset enhanced with CLAHE achieved 99% for accuracy, precision, recall, and specificity, and an mAP50-95 of 82%.

This improvement indicates that the use of CLAHE as a key component in the pre-processing method effectively enhances image feature clarity, leading to better object detection performance. Furthermore, as shown in Figure 4, the resulting IoU scores are also notably high, with values exceeding 90%.

C. *Edge Deployment and Visualization Output*

After evaluating the model, it can be observed in Figure 5 that the most effective version was implemented on the NVIDIA Jetson Orin Nano platform to facilitate real-time inference. The selected model demonstrated reliable prediction accuracy when evaluated on the edge device, characterized by low latency and optimal resource usage. This configuration enabled the system to perform efficient, on-device processing, allowing it to identify normal and defective white pepper seeds with high precision within just a few seconds after image submission. Such performance highlights the model's suitability for deployment in practical, time-sensitive agricultural applications where immediate feedback is essential.

To enhance user engagement and improve accessibility, the prediction results were integrated into a user-friendly, web-based interface developed using Streamlit. This

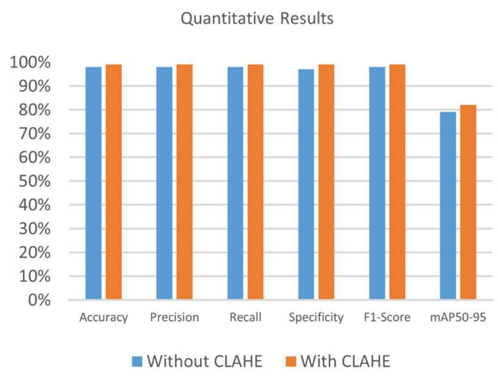
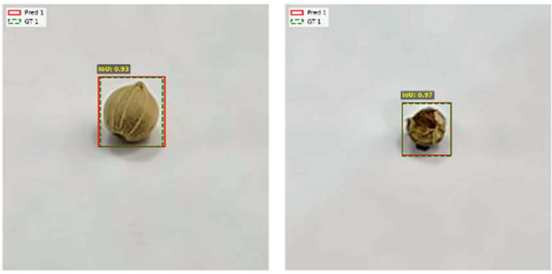


Fig. 3. Quantitative Evaluation Comparison



(a) (b)

Fig. 4. IoU Score Result (a) Normal (b) Defect

interactive interface allows users to upload images of white pepper seeds and receive real-time detection results, including annotated bounding boxes that highlight the identified objects directly on the original images. Additionally, the interface displays corresponding confidence scores for each prediction, providing insight into the model's certainty levels. Users can also monitor the inference duration, which reflects the system's processing speed on the edge device. By combining model output visualization with performance metrics in an intuitive layout, the Streamlit interface facilitates a seamless and informative user experience. A representation of this interface is shown in Figure 6.

This visualization platform guarantees that the system is readily available to a diverse user base without necessitating advanced technical skills, thereby proving beneficial for pepper seed producers in executing sorting operations, even in environments with limited resources. A demonstration video showcasing the system's capabilities can be viewed on YouTube at https://youtu.be/m5_mjJ0tKMs.

IV. Conclusion

This study presents the development and evaluation of a defect detection system for white pepper seeds based on deep learning, utilizing the YOLOv8m architecture. The use of CLAHE as a key component in the pre-processing method effectively enhances image quality, contributing to improved model performance. Evaluation results demonstrate improvements across all metrics when using CLAHE, achieving accuracy, precision, recall, specificity, and F1-score of up to 99%, and an mAP50-95 of 82%. Additionally, the IoU

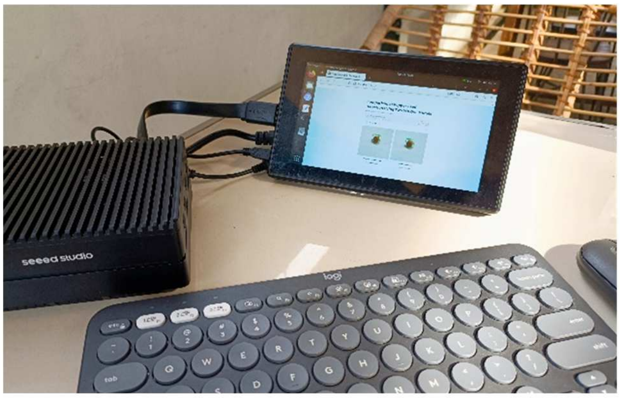


Fig. 5. Model Implementation on the NVIDIA Jetson Orin Nano Platform

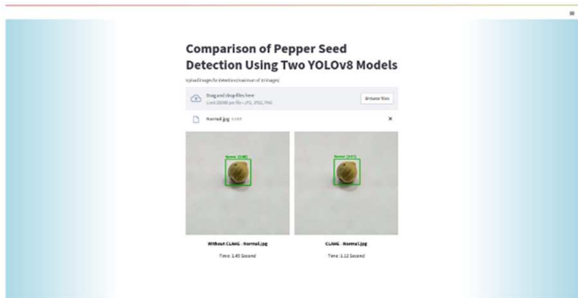


Fig. 6. Main interface of the White Pepper Defect Detector deployed on the Edge-AI Jetson Orin device

scores exceeding 90% indicate a high level of detection accuracy.

Furthermore, deploying the trained model on the NVIDIA Jetson Orin Nano demonstrates the feasibility of implementing a lightweight and efficient Edge AI system. The integration with a Streamlit-based interface also offers an accessible and interactive platform for real-time defect detection in white pepper seeds. This research highlights the effectiveness of combining image enhancement techniques with deep learning and edge deployment to support practical applications, particularly for early detection tasks.

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References

[1] F. Denaneera Waardhana and T. Sarianti, "Factors Affecting Pepper Production In Indonesia," vol. 8, no. 2, pp. 677–686, 2024, doi: 10.21776/ub.jepa.2024.008.02.23.

[2] N. Mahdi and S. Suprehatin, "Markets Position Of Indonesian Pepper In The Global Market," Jurnal Ekonomi Pertanian dan Agribisnis, vol. 5, no. 2, pp. 595–605, Apr 2021, doi: 10.21776/ub.jepa.2021.005.02.26.

[3] KARIMU ABDULAI, "Development And Evaluation Of A Semi Automatic Pepper Seedling Transplater," 2020. [Online]. Available: https://ir.ucc.edu.gh/xmlui [Accessed: Apr. 22, 2025].

[4] I. Darmawan and T. Prayogi, "Design and Simulation of a Soaking and Separation Machine for Pepper with a Capacity of 210 Kg," 2017.

[5] R. Lionnie and M. Alaydrus, "Study of the Effects of Histogram Equalization and CLAHE in Detecting Human Facial Features," *J. Tek.*

Elektro (JTE), vol. 15, no. 02, pp. 89–94, 2024, doi: 10.22441/jte.2024.v15i2.002.

[6] E. Badidi, K. Moumane, and F. El Ghazi, “Opportunities, Applications, and Challenges of Edge-AI Enabled Video Analytics in Smart Cities: A Systematic Review,” 2023, Institute of Electrical and Electronics Engineers Inc. doi: 10.1109/ACCESS.2023.3300658.

[7] Qifa, "Pepper Seeds Detection," Roboflow Universe. [Online]. Available: https://universe.roboflow.com/qifa/pepper-seeds-detection-gaed6. [Accessed: Apr. 22, 2025].

[8] Y.-T. Peng, H.-H. Liao, and C.-F. Chen, "Two-Exposure Image Fusion Based on Optimized Adaptive Gamma Correction," Sensors, vol. 22, no. 1, art. 24, Dec. 2022. doi: 10.3390/s22010024.

[9] S. Nuraisha dan S. Handayani, “Analysis of CLAHE implementation for fake fingerprint image detection,” *Djtechno: J. Inf. Technol*. Res., vol. 2, no. 1, pp. 27–34, Jul. 2021.

[10] S. Fang, J. Wu, and S. Wu, "A Content-Aware Non-Local Means Method for Image Denoising," Electronics, vol. 11, no. 18, art. 2898, Sept. 2022. doi: 10.3390/electronics11182898.

[11] Y. Song, C. Li, S. Xiao, H. Xiao, dan B. Guo, “Unsharp masking image enhancement: The parallel algorithm based on cross-platform,” Sci. Rep., vol. 12, art. 20175, pp. 1–10, 2022, doi: 10.1038/s41598-022-21745-9.

[12] J. Zophie dan H. T. Triharminto, “Implementation of the You Only Look Once (YOLO) algorithm using a web camera for detecting static and dynamic objects,” *J. TNI Angkatan Udara*, vol. 1, no. 1, pp. 15–22, 2022, doi: 10.62828/jpb.v1i1.50.

[13] Husnan, C. Fatichah, and R. Dikairono, “Object detection using the YOLO method and its implementation on an underwater robot,” *J. Tek. ITS*, vol. 12, no. 3, pp. 1–6, 2023, ISSN: 2337-3539.

[14] K. M. Azhar, I. Santoso, and Y. A. A. Soetrisno, “Implementation of Deep Learning Using the Convolutional Neural Network Method and the YOLO Algorithm in a Banknote Detection System for Individuals with Low Vision,” *Transient: Jurnal Ilmiah Teknik Elektro*, vol. 10, no. 3, hlm. 502–509, Sep 2021, doi: 10.14710/transient.v10i3.502-509.

[15] D. Nafis Alfarizi, R. Agung Pangestu, D. Aditya, M. Adi Setiawan, and P. Rosyani, “The Use of YOLO Method in Object Detection: A Systematic Literature Review,” 2023. Available: https://jurnalmahasiswa.com/index.php/aidanspk. [Accessed: Apr. 22, 2025].

[16] Yanto, F. Aziz, and Irmawati, “YOLO-V8 Algorithm Improvement for Face Mask Detection,” *JATI (Jurnal Mahasiswa Teknik Informatika)* , vol. 7, 2023. Available: https://doi.org/10.36040/jati.v7i3.7047.

[17] T. Fawcett, “ROC graphs: Notes and practical considerations for researchers,” Mach. Learn., vol. 31, no. 1, pp. 1–38, 2004.

[18] H. V. Pham, T. G. Tran, C. D. Le, A. D. Le, and H. B. Vo, “Benchmarking Jetson edge devices with an end-to-end video-based anomaly detection system,” in Advances in Information and Communication, K. Arai, Ed. Cham, Switzerland: Springer, 2024, vol. 920, pp. 358–374, doi: 10.1007/978-3-031-44067-0_25.

[19] Muchtar, Kahlil, et al. “Edge AI-Based Detection for Defective Coffee Beans using Deep Learning and Streamlit Framework,” *IEEE Access*, Apr. 2025.

[20] M. Kuzlu, F. O. Catak, S. Sarp, U. Cali, and O. Guler, "A Streamlit-based Artificial Intelligence Trust Platform for Next-Generation Wireless Networks," IEEE Access, 2025.